

# Raport Ewaluacji Modeli ADMET

## 1. Porównanie architektur w modelu Random Forest

| Endpoint / Zbiór danych   | Embeddings                                      | Fingerprints                                    | Descriptors                                     |
|---------------------------|-------------------------------------------------|-------------------------------------------------|-------------------------------------------------|
| Caco2_Wang                | RMSE: 0.5364<br>MAE: 0.4211<br>R²: 0.5497       | RMSE: 0.4968<br>MAE: 0.3771<br>R²: 0.6115       | RMSE: 0.4652<br>MAE: 0.3458<br>R²: 0.6594       |
| Lipophilicity_AstraZeneca | RMSE: 0.9681<br>MAE: 0.7673<br>R²: 0.3657       | RMSE: 0.8650<br>MAE: 0.6511<br>R²: 0.4935       | RMSE: 0.8902<br>MAE: 0.6849<br>R²: 0.4636       |
| Solubility_AqSolDB        | RMSE: 1.3995<br>MAE: 1.0813<br>R²: 0.6391       | RMSE: 1.3232<br>MAE: 0.9851<br>R²: 0.6774       | RMSE: 1.1236<br>MAE: 0.7909<br>R²: 0.7673       |
| HIA_Hou                   | Accuracy: 0.9052<br>F1: 0.9463<br>AUROC: 0.9832 | Accuracy: 0.8793<br>F1: 0.9327<br>AUROC: 0.8882 | Accuracy: 0.9138<br>F1: 0.9510<br>AUROC: 0.8890 |
| Half_Life_Obach           | RMSE: 114.9563<br>MAE: 32.1374<br>R²: 0.0718    | RMSE: 101.7780<br>MAE: 26.7299<br>R²: 0.2778    | RMSE: 111.1411<br>MAE: 26.3643<br>R²: 0.1389    |
| Clearance_Hepatocyte_AZ   | RMSE: 37.8851<br>MAE: 27.6711<br>R²: 0.4071     | RMSE: 45.9328<br>MAE: 34.4996<br>R²: 0.1548     | RMSE: 47.4932<br>MAE: 37.2552<br>R²: 0.0964     |
| CYP3A4_Veith              | Accuracy: 0.7733<br>F1: 0.6950<br>AUROC: 0.8487 | Accuracy: 0.7928<br>F1: 0.7257<br>AUROC: 0.8794 | Accuracy: 0.7393<br>F1: 0.7210<br>AUROC: 0.8172 |
| VDss_Lombardo             | RMSE: 6.8932<br>MAE: 4.5002<br>R²: 0.0014       | RMSE: 5.7826<br>MAE: 3.6000<br>R²: 0.2837       | RMSE: 7.4404<br>MAE: 3.8296<br>R²: -0.1859      |
| AMES                      | Accuracy: 0.8048<br>F1: 0.8245<br>AUROC: 0.8760 | Accuracy: 0.8407<br>F1: 0.8541<br>AUROC: 0.9154 | Accuracy: 0.7761<br>F1: 0.7998<br>AUROC: 0.8659 |
| hERG                      | Accuracy: 0.8722<br>F1: 0.9202<br>AUROC: 0.9030 | Accuracy: 0.8473<br>F1: 0.9038<br>AUROC: 0.8143 | Accuracy: 0.8168<br>F1: 0.8776<br>AUROC: 0.8326 |

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## 2. Porównanie architektur w Sieciach Neuronowych (NN)

| Endpoint / Zbiór danych   | Embeddings                                      | Fingerprints                                    | Descriptors                                     |
|---------------------------|-------------------------------------------------|-------------------------------------------------|-------------------------------------------------|
| Caco2_Wang                | RMSE: 0.4768<br>MAE: 0.3720<br>R²: 0.6442       | RMSE: 0.4942<br>MAE: 0.3762<br>R²: 0.6155       | RMSE: 0.4763<br>MAE: 0.3369<br>R²: 0.6428       |
| Lipophilicity_AstraZeneca | RMSE: 0.7370<br>MAE: 0.5725<br>R²: 0.6323       | RMSE: 0.8067<br>MAE: 0.5922<br>R²: 0.5596       | RMSE: 0.8958<br>MAE: 0.6804<br>R²: 0.4568       |
| Solubility_AqSolDB        | RMSE: 1.1068<br>MAE: 0.8094<br>R²: 0.7743       | RMSE: 1.2635<br>MAE: 0.9272<br>R²: 0.7058       | RMSE: 1.2061<br>MAE: 0.8663<br>R²: 0.7319       |
| HIA_Hou                   | Accuracy: 0.9569<br>F1: 0.9741<br>AUROC: 0.9881 | Accuracy: 0.9224<br>F1: 0.9557<br>AUROC: 0.8649 | Accuracy: 0.9052<br>F1: 0.9453<br>AUROC: 0.9300 |
| Half_Life_Obach           | RMSE: 93.5280<br>MAE: 26.9926<br>R²: 0.3856     | RMSE: 85.1543<br>MAE: 24.9387<br>R²: 0.4945     | RMSE: 98.7200<br>MAE: 27.7646<br>R²: 0.3206     |
| Clearance_Hepatocyte_AZ   | RMSE: 36.5755<br>MAE: 23.1591<br>R²: 0.4474     | RMSE: 49.0328<br>MAE: 35.0261<br>R²: 0.0368     | RMSE: 47.2170<br>MAE: 34.9098<br>R²: 0.1069     |
| CYP3A4_Veith              | Accuracy: 0.8001<br>F1: 0.7509<br>AUROC: 0.8788 | Accuracy: 0.7985<br>F1: 0.7421<br>AUROC: 0.8757 | Accuracy: 0.7453<br>F1: 0.6937<br>AUROC: 0.8236 |
| VDss_Lombardo             | RMSE: 7.5154<br>MAE: 3.9647<br>R²: -0.1870      | RMSE: 5.8750<br>MAE: 3.5219<br>R²: 0.2606       | RMSE: 11.1298<br>MAE: 5.6490<br>R²: -1.6535     |
| AMES                      | Accuracy: 0.8061<br>F1: 0.8260<br>AUROC: 0.8812 | Accuracy: 0.8091<br>F1: 0.8263<br>AUROC: 0.8878 | Accuracy: 0.7321<br>F1: 0.7464<br>AUROC: 0.8085 |
| hERG                      | Accuracy: 0.8496<br>F1: 0.9010<br>AUROC: 0.8692 | Accuracy: 0.7481<br>F1: 0.8421<br>AUROC: 0.7948 | Accuracy: 0.8321<br>F1: 0.8911<br>AUROC: 0.7883 |